Recitation 11

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Outline

1. **Review**

2. **Recitation**

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Review

- Sum of independent RVs

$$M_Z(s) = M_{X_1}(s) \cdots M_{X_n}(s)$$

- Sum of a random number of independent RVs

$$\mathbf{E}[Y] = \mathbf{E}[N]\,\mathbf{E}[X]$$

$$\mathrm{var}(Y) = \mathbf{E}[N]\,\mathrm{var}(X) + (\mathbf{E}[X])^2\,\mathrm{var}(N)$$

$$M_Y(s) = \sum_{n=0}^{\infty} (M_X(s))^n p_N(n) = M_N(\log M_X(s))$$

1. Review

2. **Recitation**

1. You are visiting the rainforest, but unfortunately your insect repellent has run out. As a result, at each second, a mosquito lands on your neck with probability 0.5. If one lands, with probability 0.2 it bites you, and with probability 0.8 it never bothers you, independently of other mosquitoes.

   (a) What is the expected time between successive mosquito bites? What is the variance of the time between successive mosquito bites?

   (b) In addition, a tick lands on your neck with probability 0.1. If one lands, with probability 0.7 it bites you, and with probability 0.3, it never bothers you, independently of other ticks and mosquitoes. Now, what is expected time between successive bug bites? What is the variance of the time between successive bug bites?

(a) Let X denote the time between successive mosquito bites. The mosquito bites with probability $p = 0.5 \times 0.2 = 0.1$. Therefore, X follows a geometric distribution with parameter $p = 0.1$.

$$\mathbf{E}[X] = \frac{1}{p} = 10$$

$$\mathrm{var}(X) = \frac{1-p}{p^2} = 90$$

(b) Tick bites occur according to another Bernoulli process with parameter $q = 0.1 \times 0.7 = 0.07$. Bug bites occur according to a merged Bernoulli process with parameter $r = p+q-pq = 0.163$. Let Y = time between successive bug bites, so

$$\mathbf{E}[Y] = \frac{1}{r} \approx 6.135$$

$$\mathrm{var}(Y) = \frac{1-r}{r^2} \approx 31.503$$

2. Suppose there are n papers in a drawer. You draw a paper and sign it, and then, instead of filing it away, you place the paper back into the drawer. If any paper is equally likely to be drawn each time, independent of all other draws, what is the expected number of papers that you will draw before signing all n papers?

Let M be the total number of draws you make until you have signed all n papers. Let T_i be the number of draws you make until drawing the next unsigned paper after having signed i papers. Then $M = T_0 + \cdots + T_{n-1}$.

View the process of selecting the next unsigned paper after having signed i papers as a sequence of independent Bernoulli trials with probability of success $p_i = \frac{n-i}{n}$. So,

$$\mathbf{E}[T_i] = \frac{n}{n-i}$$

and the expected number of papers that you will draw before signing all n papers

$$\mathbf{E}[M] = \sum_{i=0}^{n-1} \mathbf{E}[T_i] = n \sum_{k=1}^{n} \frac{1}{k}.$$

3. Show that the sum of a Poisson-distributed number of i.i.d. Bernoulli random variables is Poisson.

Hint: The transform of Poisson RV with parameter λ is $M_X(s) = e^{\lambda(e^s-1)}$, and the transform of Bernoulli RV with parameter p is $M_X(s) = 1 - p + pe^s$.

Let N be a Poisson-distributed RV with parameter λ. Let $X_i, i = 1, \cdots, N$, be i.i.d. Bernoulli RVs with parameter p, and let $L = X_1 + \cdots + X_N$ be the corresponding sum.
Using transform, we have

$$M_N(s) = e^{\lambda(e^s - 1)}$$

$$M_X(s) = 1 - p + pe^s$$

Therefore,

$$\begin{aligned} M_L(s) = M_N(\log M_X(s)) &= e^{\lambda(M_X(s) - 1)} \\ &= e^{\lambda(-p + pe^s)} \\ &= e^{\lambda p(e^s - 1)} \end{aligned}$$

L is a Poisson RV with parameter λp.

4. The color of a person's eyes is determined by a single pair of genes. If they are both blue-eyed genes, then the person will have blue eyes; if they are both brown-eyed genes, then the person will have brown eyes; and if one of them is a blue-eyed gene and the other a brown-eyed gene, then the person will have brown eyes. (Because of the latter fact, we say that the brown-eyed gene is *dominant* over the blue-eyed one.) A newborn child independently receives one eye gene from each of its parents, and the gene it receives from a parent is equally likely to be either of the two eye genes of that parent. Suppose that Smith and both of his parents have brown eyes, but Smith's sister has blue eyes.

   (a) What is the probability that Smith possesses a blue-eyed gene?
   (b) Suppose that Smith's wife has blue eyes. What is the probability that their first child will have blue eyes?
   (c) Based on (b), if their first child has brown eyes, what is the probability that their next child will also have brown eyes?

5. There is a circuit shown below. Three switches S_1, S_2, S_3 are all turned on. The lifetime of S_1 follows uniform distribution between 0 and 4 minutes. The lifetime of S_2 and S_3 follow exponential distribution, with ***mean*** 3 minutes and 6 minutes respectively. All the other components never break down.

(a) Find the PDF of the lighting time of light L_1.

(b) Find the PDF of the lighting time of light L_2.

(c) What is the expectation of the lighting time of light L_2?